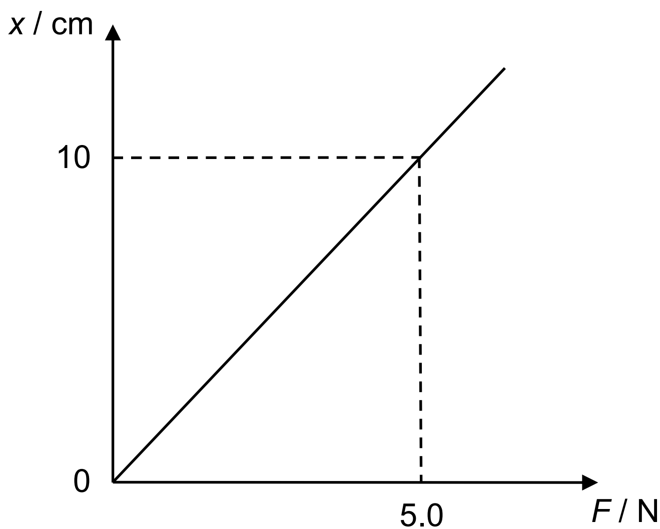
7 The variation of the extension x of a light spring with the force F applied is shown below.



Slotted masses with a total weight of 3.0 N were initially hung on the spring.

What is the decrease in the elastic potential energy stored in the spring when a slotted mass weighing 0.5 N is removed from the system?

A	0.013 J	**B**	0.015 J	**C**	0.028 J	**D**	0.056 J

Answer: C

Force constant, $k = \dfrac{5.0}{0.1} = 50 \text{ N m}^{-1}$

Initial extension $= \dfrac{F}{k} = \dfrac{3}{50} = 0.06 \text{ m}$ and final extension $= \dfrac{F}{k} = \dfrac{2.5}{50} = 0.05 \text{ m}$

Change in E.P.E. $= \dfrac{1}{2}kx_f^2 - \dfrac{1}{2}kx_i^2 = \dfrac{1}{2}(50)(0.05)^2 - \dfrac{1}{2}(50)(0.06)^2 = -0.028 \text{ J}$

8	A block is pulled with a constant force of 25 N on a rough horizontal surface. It accelerates